Design and Development of Conveyor Drive System and Gradient Drying in a Battery Production Line

Project Review

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Seminar Outline

1. Introduction
   - Introduction
   - Problem Statement
2. Proposed Work
   - Work Methodology
   - Work Done
3. Conclusions and Future Scope



Introduction

- Thin-film deposition is essential in the development of several industries in the modern world, including electronics, energy storage, healthcare, and aerospace.
- One of the most important applications of thin-film is battery technology.
- Slot-die coating is most commonly used thin-film deposition technique due to its properties such as higher precision and scalability for large-scale battery manufacturing lines.

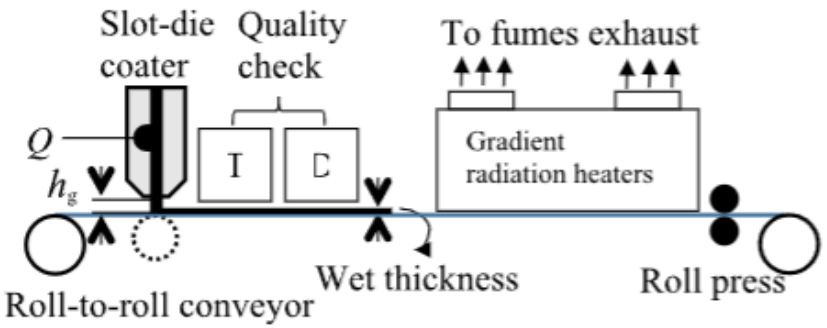


Fig 1. The schematic diagram of battery electrode coating system.

Motivation

- The primary objective is to ensure a uniform, defect-free coating throughout the process.
- Speed control in conveyor systems is critical in battery manufacturing lines, such as the slot-die coating process.
- Gradient temperature is crucial for controlled drying and high-quality thin film formation. Here's the significance of each stage:

Stage	Temperature (°C)	Key function	Reason
Stage 1	30–40	Initial evaporation	Prevent surface skin formation, allow flow leveling and uniform nucleation
Stage 2	40–50	Intermediate drying	Control phase separation, stabilize morphology, improve surface smoothness
Stage 3	50–60	Final drying and curing	Remove residual solvent, enhance film properties, prepare for next processing steps

Table: Drying stages and their significance in thin-film coating

Problem Statement

- To design a closed-loop control system to control the speed of the conveyor system.
- To obtain a consistent conveyor speed without any fluctuations for a uniform defect-free thin-film wet coat.
- To maintain constant temperature at each stage (3 stages) of the coating process using a closed-loop system.



Proposed Control System

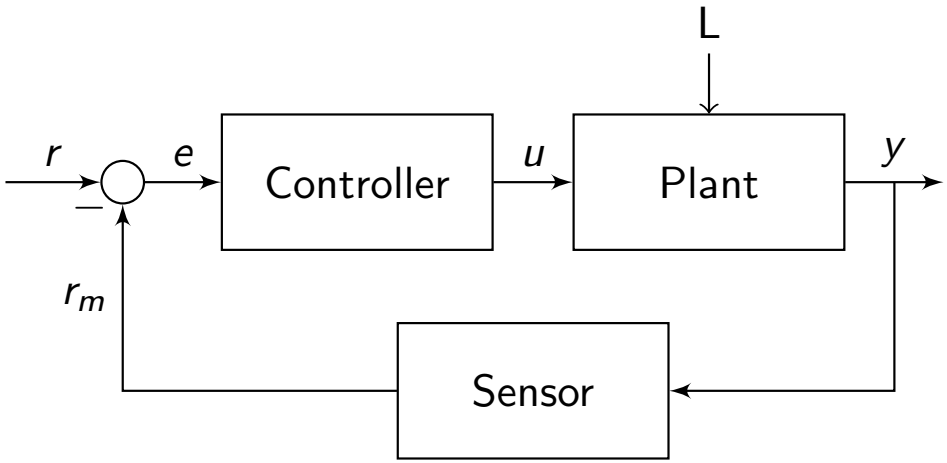


Figure: The schematic diagram of closed-loop control system.

where :

- r = Target rpm / temperature
- r_m = Measured rpm / temperature
- e = r - r_m
- u = Control signal
- L = External load
- y = Output

Experimental Setup of Conveyor Drive System

- An Proportional-Integral (PI) control algorithm is used in the Arduino microcontroller.
- RPM is measured using an infrared sensor that is fed back to the controller making it a closed-loop system.

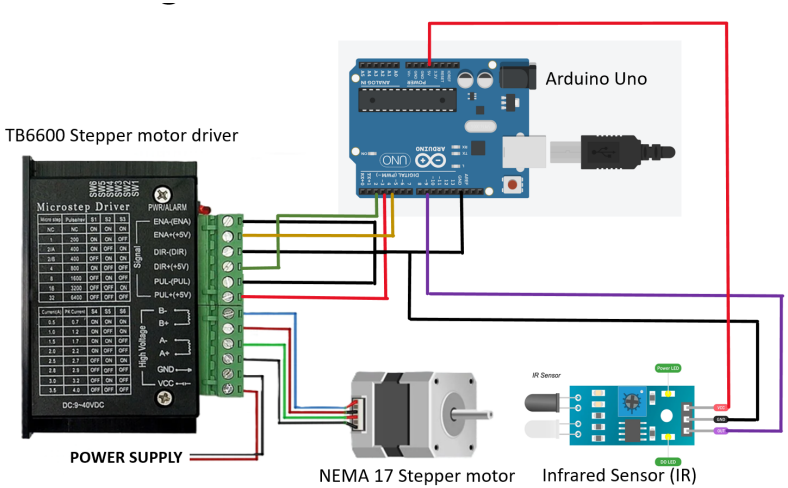


Figure: Connection diagram of conveyor drive system.

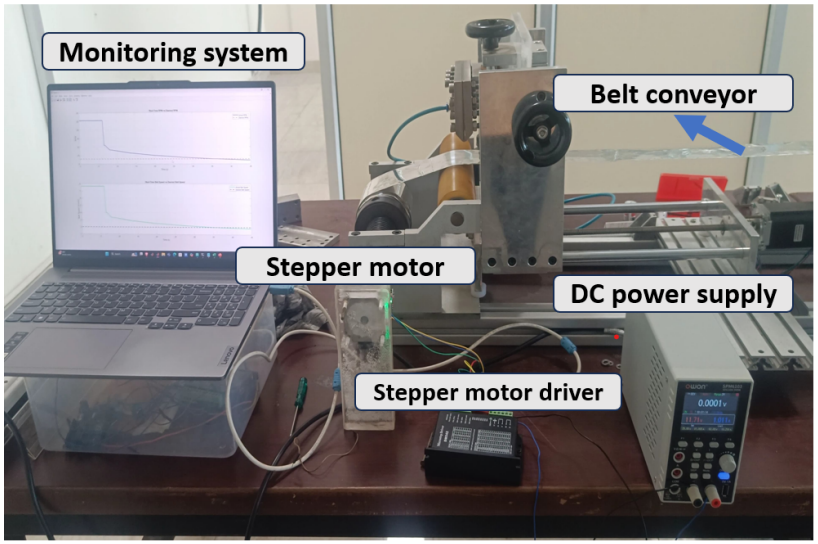


Figure: Experimental setup of conveyor drive system.

Experimental Setup of Temperature Control System

- Used a 34970A Data Acquisition by Agilent for logging the temperature data for observing the controller's behavior for various temperatures.

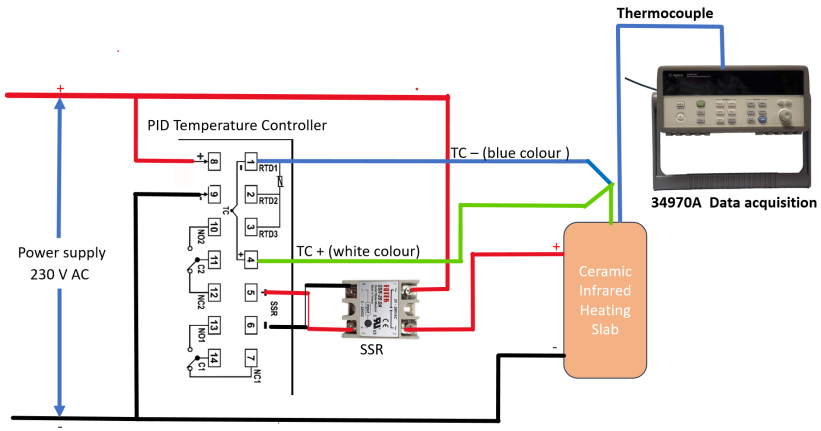


Figure: Connection diagram of temperature control system.

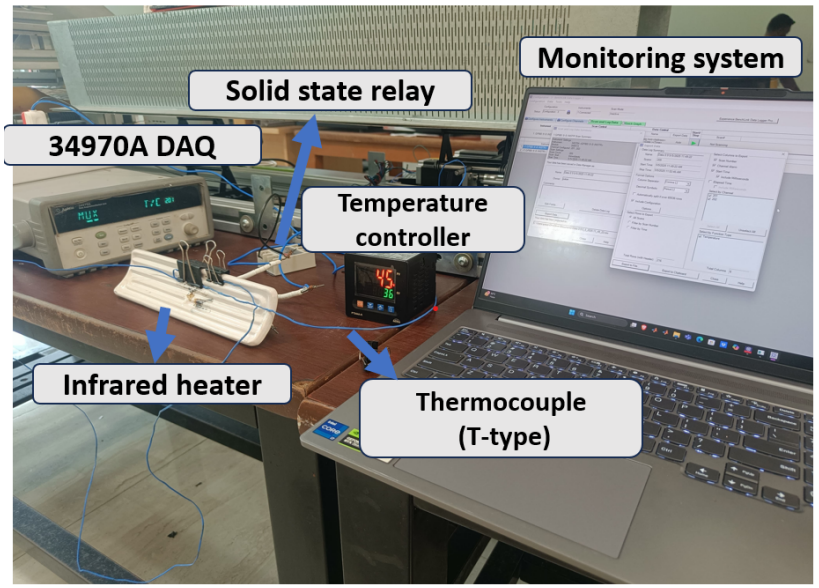


Figure: Experimental setup of temperature control system.

Hardware Setup

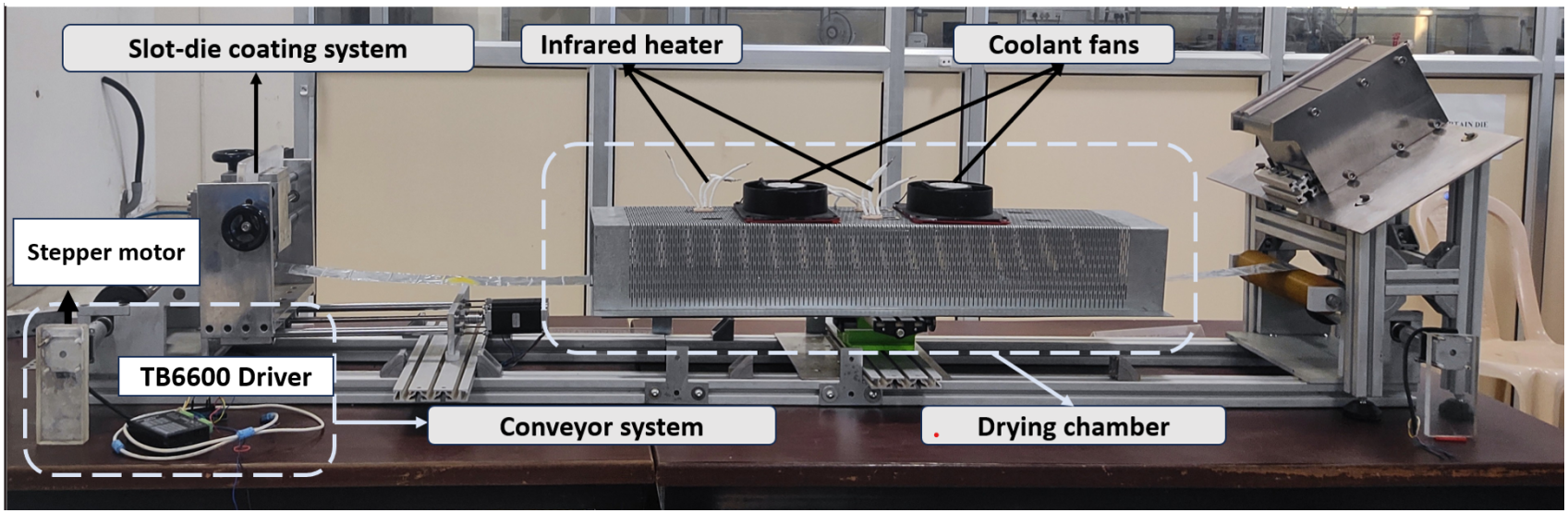


Figure: Hardware setup developed for battery in-line manufacturing.

- The hardware setup consists of a slot-die coating system integrated with a conveyor-driven drying chamber. A NEMA 17 stepper motor, controlled via a TB6600 driver, moves the coated substrate under infrared heaters and coolant fans to ensure uniform drying.

Components and Software Used

Category	Component / Device	Purpose
Microcontroller	Arduino Uno	Real-time control of stepper motor and speed feedback
Motor	NEMA 23 Stepper Motor	Drives the conveyor belt
Motor Driver	TB6600 Stepper Motor Driver	Controls current and step signals to motor
Sensor	IR Sensor	Measures motor speed for feedback loop
Controller	PT344A-K Temperature Controller	Regulates IR heater temperature via SSR
Heater	Infrared Heating Slabs	Provides controlled heating for slurry drying
Switching Device	Solid State Relay (SSR)	Turns heating slab on/off based on controller signal
Temperature Sensor	T-type Thermocouple	Provides feedback to temperature controller
Data Acquisition	34970A DAQ Unit	Acquires real-time temperature data for logging
Power Supply	24V DC Supply & 230V AC Line	Powers motor and heating elements

Software Used	Purpose
Arduino IDE	Coding and uploading PI controller to Arduino
MATLAB & Simulink	Modeling, controller tuning, and performance validation
Keysight Connection Expert	Establishing communication with measurement instruments (34970A DAQ)
Agilent BenchLink Data Logger	Real-time data logging of temperature



Results of Conveyor Speed Control

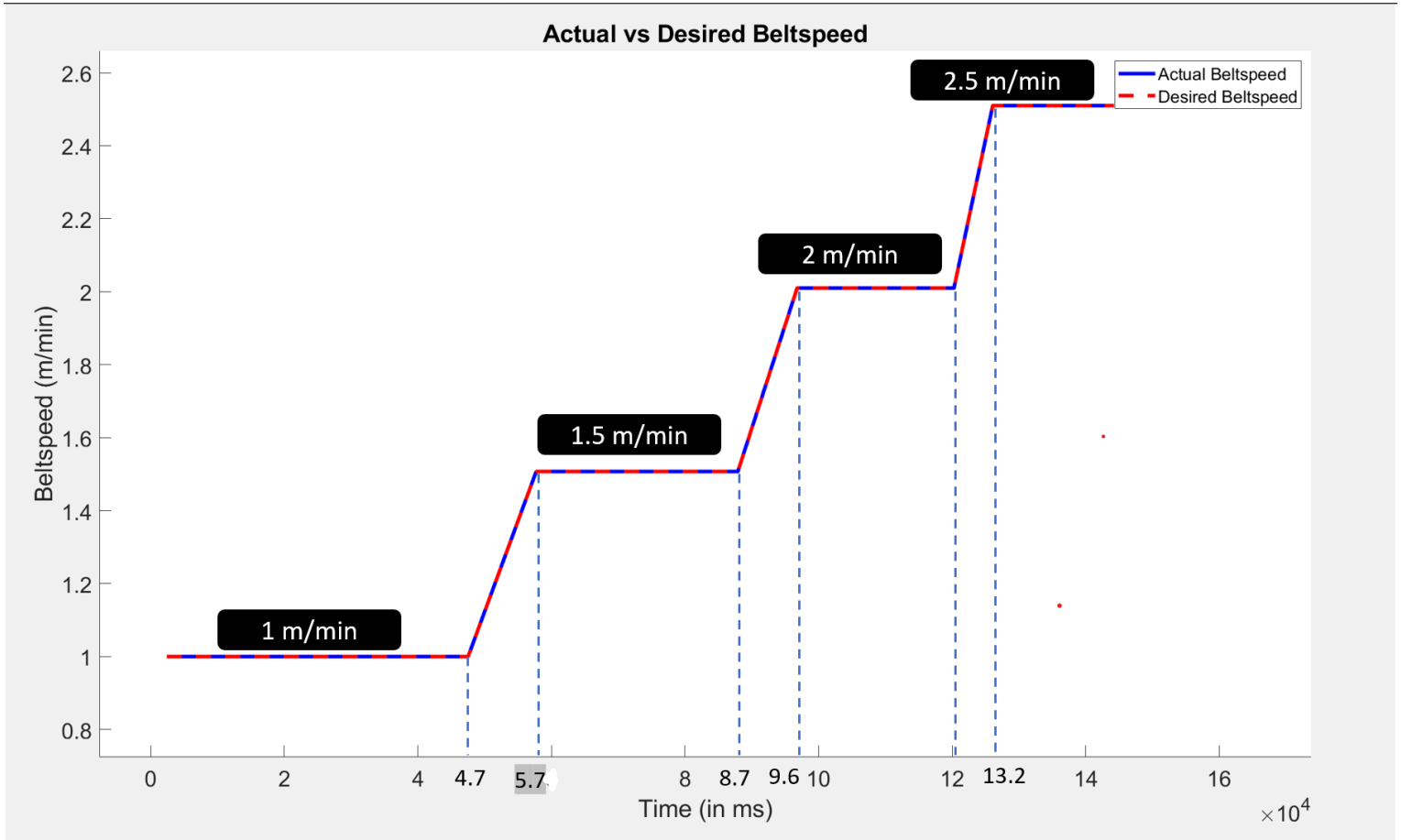
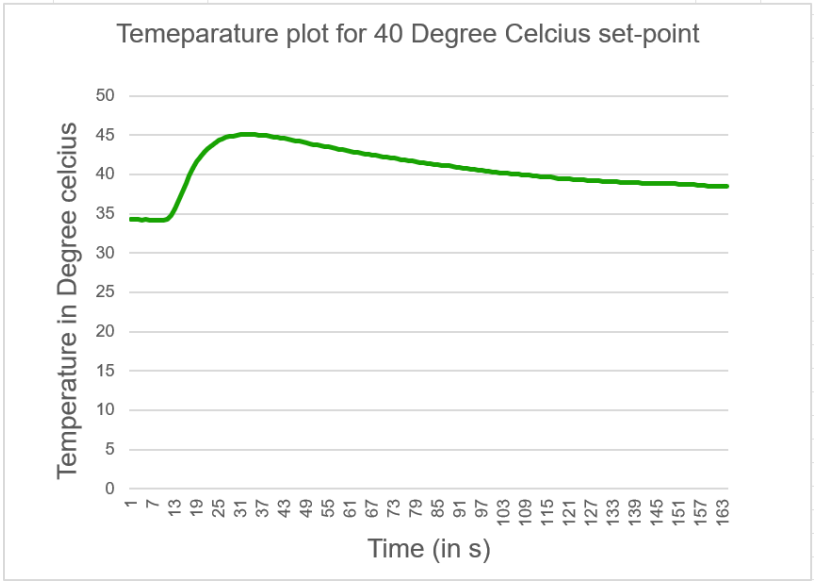


Figure: Behavior of the conveyor drive system for sudden change in the speeds(1 m/min, 1.5 m/min, 2 m/min and 2.5 m/min.

Results of Temperature Logging I



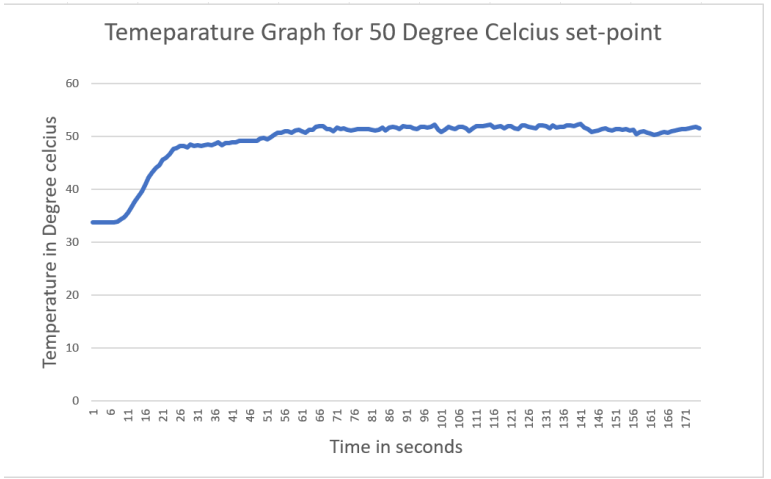


Figure: Temperature distribution plots at (left) 40°C and (right) 50°C.

Results of Temperature Logging II

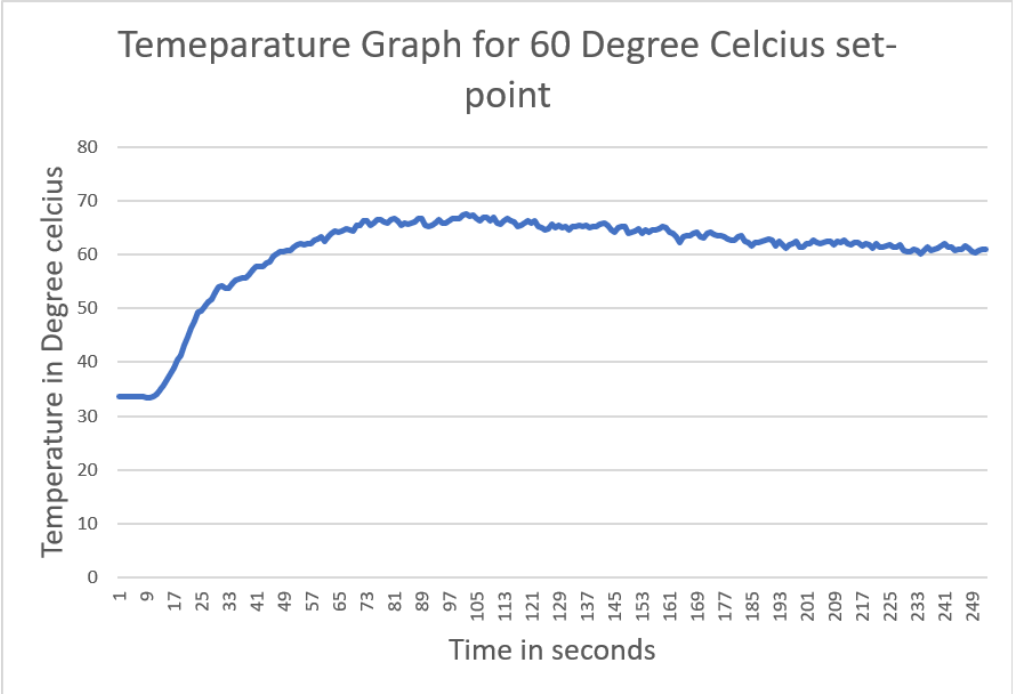


Figure: Temperature distribution plots at 60°C

- Similar plots are observed from the data collected manually from temperature controller.

Conclusions

- In the present work, the PI algorithm method is proposed to regulate the speed of the conveyor belt in the battery manufacturing system.
- Obtained a stable conveyor operation at different speeds of 1 m/min, 1.5 m/min, 2 m/min, 2.5 m/min, 3 m/min, 3.5 m/min, 4 m/min, 4.5 m/min and 5 m/min.
- It was observed that sudden changes in speed did not occur instantly due to a noticeable delay caused by the time taken for the motor shaft to interrupt the IR sensor, thereby affecting the responsiveness of the feedback loop.
- Tuned gain values (Kp, Ki and Kd) of the temperature controller with the infrared heating slab and data logging have been performed using 34970A DAQ.
- Settling time ($\pm 2\% error$) for 40 °C, 50 °C, 60 °C are 93 s, 48 s, 140 s respectively.

Future Scope

- The same setup of conveyor drive system will be placed at the other end of the in-line manufacturing system to ensure enough web tension to obtain a defect-free thin-film coat.
- To achieve a defect-free and crack-free coating after drying, gradient (three-stage) drying process will be employed.
- Integrating these proposed system with a syringe mechanism for a continuous supply of controlled slurry to the slot-die coater.

Selected References I

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Link to experimental data.


Thank You